Weekly Rapport — Week 1

Real-Time Anomaly Detection Platform for Branch Teller Operations

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Contents

Project Context and Positioning

Institution: Amen Bank — Direction Centrale du Syst‘eme d’Information
Supervisor: D'epartement des projets agence
Scope: Real-time anomaly detection on four branch teller operations (*retrait*, *versement*, *virement*, *remise de ch‘eques*)
Regulatory umbrella: LAB/FT (anti-money laundering / terrorism financing)

Amen Bank already operates an AI-based AML system performing bank-wide compliance reporting on transaction streams (audience: compliance officer; tempo: quarterly; output: CTAF / regulatory). This project complements that system with **near-real-time decision support for the branch supervisor**, focused on teller operations, incorporating the employee dimension via a dual-actor model, and using an LSTM sequence-aware architecture. Same regulatory umbrella, different tools.

This rapport covers the foundational design phase: architecture, data schemas, synthetic data strategy, and the first three source extractions.

System Architecture

The platform uses a streaming-first architecture with four microservices chained via RedPanda topics, supported by a hot/cold data layer and a batch pipeline for profile rebuilding and model retraining.

Key architectural choices

RedPanda over Kafka. Kafka-compatible API with a lighter runtime (no JVM, no ZooKeeper). Sufficient for the project's throughput requirements while reducing operational complexity.

Dual-actor model. Both the client and the teller employee are independently profiled. Each has a hot profile (Redis) for real-time lookups and a cold profile (PostgreSQL) for historical analysis and batch computation. Anomalies can originate from either actor.

SHAP co-located in the Scoring Service. The explainability module runs inside the scoring service, so the SHAP explanation travels as part of the enriched payload all the way to the `decisions` topic. This simplifies the Compliance Dashboard, which consumes a single topic containing both the alert and its explanation.

Supervisor feedback loop. The dashboard allows the supervisor to confirm or reject each alert. These decisions flow back into the batch pipeline and feed into the weekly model retraining cycle, creating a closed learning loop.

Alert severity tiers. Four levels: INFO (logged, no action), REVIEW (supervisor reviews within 24h), ALERT (immediate supervisor attention), BLOCK (automatic hold, requires override).

Streaming topology

Four RedPanda topics chained, all partitioned by `client_id` to preserve cross-operation temporal ordering for a given client (critical for the LSTM):

```
raw-events  -->  enriched-events  -->  scored-events  -->  decisions
```

Batch pipeline

Two periodic jobs:

- **Nightly profile rebuild:** reads from the `transactions` cold table, recomputes all aggregate statistics and distributions for both client and employee profiles, writes snapshots to PostgreSQL and refreshes Redis hot profiles.
- **Weekly model retraining:** retrains the Autoencoder and LSTM using the latest transaction data and supervisor feedback (confirmed/rejected alerts), logs experiments to MLflow, and promotes the best model to the scoring service.

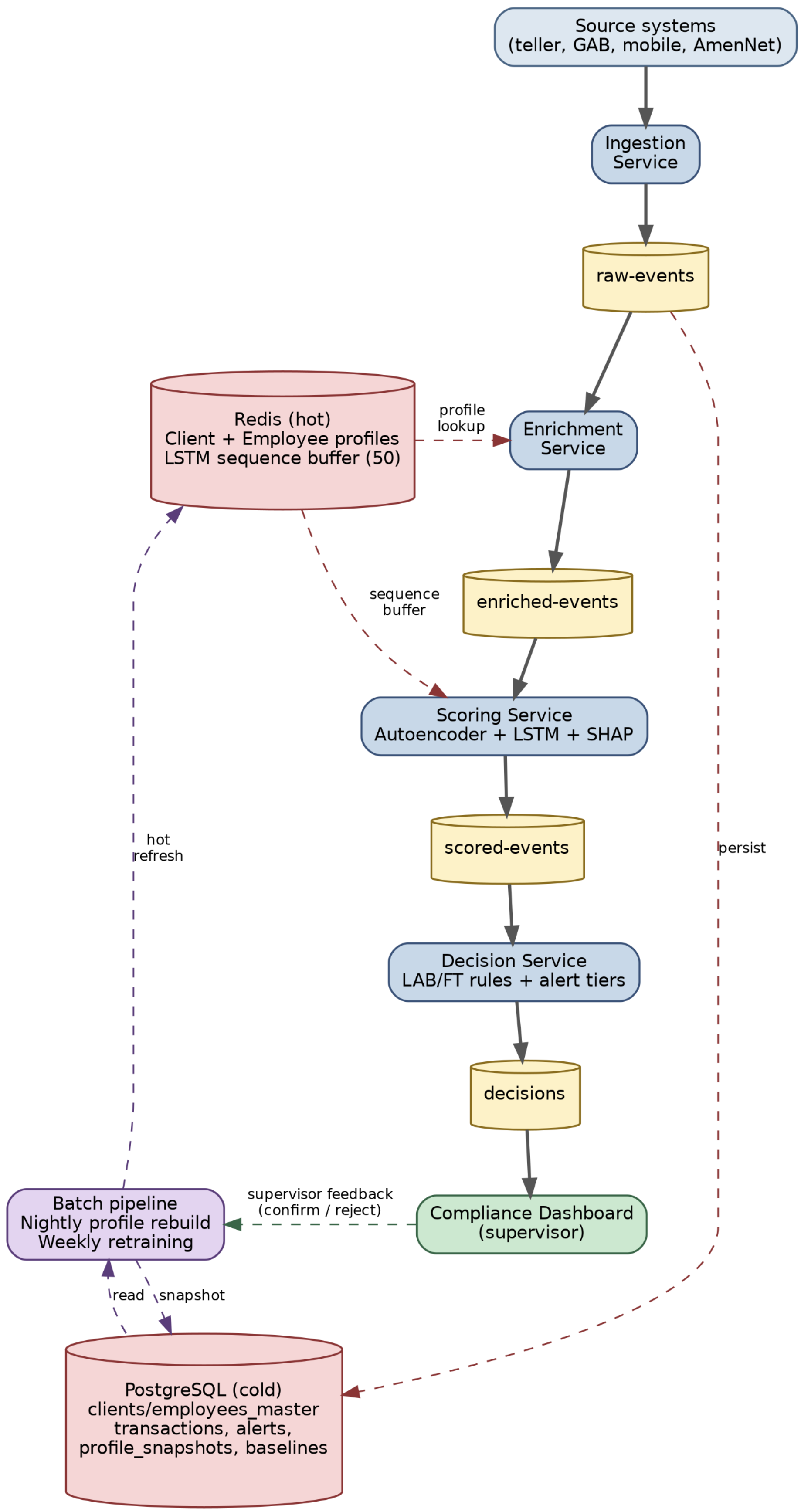

Figure 1: System Architecture

Data Schemas

Raw Event Schema

The raw event uses a **tagged-union envelope**: a common envelope of 11 fields shared by all operations, plus a `payload` field containing operation-specific data. All four operations travel on a single `raw-events` topic.

Two founding principles:

1. Raw events contain **only facts observable at transaction time** — no aggregate statistics (those belong in profiles).
2. **No derived or lagging fields** — for example, a check's bounced status is only known after clearing and belongs to a separate downstream event.

Common envelope

```
raw_event {
  event_id        : string   (UUID)
  client_id       : string   (FK)
  account_id      : string   (FK)
  employee_id     : string   (FK)
  branch_id       : string   (FK)
  timestamp       : datetime (UTC)
  amount          : float
  currency        : string   (ISO 4217, typically TND)
  channel         : enum     [guichet, GAB, mobile, ...]
  operation_type  : enum     [retrait, versement, virement, cheque]
  payload         : object   (operation-specific, see below)
}
```

Operation-specific payloads

```
payload_retrait {
  id_verified      : bool
  is_third_party   : bool
  third_party_id   : string  (nullable)
}

payload_versement {
  is_third_party_depositor : bool
  depositor_id             : string  (nullable -- required if
                                      depositor != holder)
  source_of_funds          : string
  source_verified          : bool
}

payload_virement {
  beneficiary_account   : string
  beneficiary_name      : string
```

```
  transfer_motif        : string
  is_internal_transfer  : bool
}

payload_cheque {
  remise_id                 : string  (shared across checks
                                       in the same remise)
  check_number              : string
  emitter_account           : string
  emitter_name              : string
  emitter_bank              : string
  check_date                : date
  is_endorsed_to_depositor  : bool
}
```

Design note on remise de ch‘eques: each check generates its own event (not one event per remise). The `remise_id` groups checks deposited together. This preserves per-check granularity for anomaly scoring while maintaining the grouping context.

Client Profile Schema (Hot — Redis)

Six categories, approximately 194 fields total:

Category 1 — Personal Info (7 fields)

```
client_id              : string  (PK)
archetype              : enum    [student, salaried, small_biz,
                                  retiree, big_biz]
maturity_status        : enum    [cold_start, mature]
home_branch_id         : string
client_type            : enum    [individual, business]
nationality            : string
account_opening_date   : date
```

Category 2 — Transaction Statistics (113 fields)

One global field plus a structured block per operation type per time window:

```
days_since_last_transaction : int

// For each combination of:
//   op_type in {retrait, versement, virement, cheque}
//   window  in {24h, 7d, 30d, 90d}
stats_<op>_<window> {
  count              : int
  sum_amount          : float
  mean_amount         : float
  std_amount          : float
  rejected_count      : int
  reversed_count      : int
  near_threshold_ratio : float   // fraction in 8k-9.999k DT band
}
// 4 ops x 4 windows x 7 metrics = 112 fields + 1 global = 113
```

Category 3 — Behavioral Patterns (12 structures)

Five frequency distributions computed at two horizons (30d and 180d):

```
distribution_branch_30d     : map<branch_id, count>
distribution_hour_30d       : map<hour_bucket, count>
distribution_dow_30d        : map<day_of_week, count>
distribution_op_type_30d    : map<op_type, count>
distribution_employee_30d   : map<employee_id, count>

distribution_branch_180d    : map<branch_id, count>
distribution_hour_180d      : map<hour_bucket, count>
distribution_dow_180d       : map<day_of_week, count>
distribution_op_type_180d   : map<op_type, count>
distribution_employee_180d  : map<employee_id, count>
```

Plus two relationship-tracking maps:

```
known_beneficiaries_365d : map<beneficiary_id, {count, last_seen}>
known_emitters_365d      : map<emitter_id, {count, last_seen}>
```

Category 4 — Financial State (12 fields)

```
current_balance          : float
total_debt               : float
account_count            : int
balance_mean_30d         : float
balance_std_30d          : float
balance_min_30d          : float
balance_max_30d          : float
inflow_sum_30d           : float
outflow_sum_30d          : float
inflow_sum_90d           : float
outflow_sum_90d          : float
net_flow_30d             : float   // inflow - outflow
```

Category 5 — Risk History (49 fields)

```
days_since_last_alert : int

// For each combination of:
//   tier   in {INFO, REVIEW, ALERT, BLOCK}
//   window in {30d, 90d, 180d}
risk_<tier>_<window> {
  total_count      : int
  confirmed_count  : int   // supervisor confirmed
  rejected_count   : int   // supervisor rejected
  pending_count    : int   // not yet reviewed
}
// 4 tiers x 3 windows x 4 metrics = 48 + 1 global = 49
```

Category 6 — LSTM Sequence Buffer

```
recent_events_buffer : list  // Bounded Redis list (LPUSH + LTRIM)
                             // Last 50 events for this client
                             // Directly feeds the LSTM at scoring time
```

Sequence length locked at 50 — matches the training sequence length to prevent any train/inference context mismatch.

Employee Profile Schema (Hot — Redis)

Five categories, approximately 188 fields total. Mirrors the client profile with two key differences: (1) financial state is intentionally excluded, and (2) a **peer comparison** category is added.

Category 1 — Personal Info (6 fields)

```
employee_id         : string  (PK)
branch_id           : string
role                : string
hire_date           : date
last_transfer_date  : date    // for transfer-sensitive cold-start
maturity_status     : enum    [cold_start, mature]
```

Category 2 — Transaction Statistics (113 fields)

Same structure as the client profile, but counts transactions **processed** by this employee (not transactions belonging to them).

Category 3 — Behavioral Patterns (10 distributions)

```
// Same 5 distributions at 30d and 180d, but:
//   - "client_distribution" replaces "employee_distribution"
//     (tracks which clients this teller serves most often)
//   - "branch_distribution" retained for part-time / multi-branch
```

Category 4 — Peer Comparison (10 fields, new)

Five z-scores computed at two horizons (30d and 90d), comparing this employee against the baseline of all employees in the same role at the same branch:

```
z_volume_30d                 : float
z_error_rate_30d             : float
z_near_threshold_ratio_30d   : float  // smurfing proximity
z_client_concentration_30d   : float  // serving few vs many clients
z_client_location_mismatch_30d : float

z_volume_90d                 : float
z_error_rate_90d             : float
z_near_threshold_ratio_90d   : float
z_client_concentration_90d   : float
z_client_location_mismatch_90d : float
```

Category 5 — Risk History

Same structure as client profile risk history.

Financial state is excluded — monitoring employee personal finances is out of scope and raises ethical concerns.

Enriched Event Schema

The enriched event is the **feature vector consumed by the Scoring Service**. The Enrichment Service constructs it by joining the raw event with profile lookups (client + employee) and computing derived features. Approximately **64 derived features across 12 sections** (section 12 added during CTAF extraction):

#	Section	Content
1	Metadata / traceability	event_id, enrichment_timestamp, profile_version
2	Raw event passthrough	All envelope fields forwarded
3	Cold-start flags	client_is_cold_start, employee_is_cold_start
4	Regulatory rule signals	Boolean flags (is_third_party_deposit, is_near_threshold, is_ppe_to_ppe_flow, etc.)
5	Client-side derived features	z-scores against archetype baseline, frequency lookups, cumulative amounts, known beneficiary/emitter checks
6	Archetype baseline features	Expected values for cold-start fallback
7	Behavioral drift features	JS-divergence between 30d and 180d distributions
8	Employee-side derived features	z-scores for the processing teller
9	Peer comparison passthrough	Employee z-scores from hot profile
10	Risk history signals	Recent alert counts and confirmation rates
11	LSTM sequence buffer	Last 50 events for the client
12	Counterparty-side derived features	Archetype, sector, flags of the counterparty when internal to bank (added during CTAF case extraction)

Cold Store Tables (PostgreSQL)

Client-side

Table	Purpose
`clients_master`	Client KYC data, archetype assignment, PPE flag, multi-role flag, controlled_accounts_list
`accounts`	Account metadata: account_id, client_id, currency, account_type (personal/corporate/self_employed), sector_code
`account_balances`	Daily balance snapshots
`transactions`	Persisted raw events (write-through from the raw-events topic)
`alerts`	All generated alerts with tier, score, SHAP explanation, supervisor decision
`profile_snapshots`	Periodic snapshots of the full client hot profile for audit and retraining
`archetype_baselines`	Expected mean/std per metric, segmented by archetype × sector × flag-set × window

Employee-side

Table	Purpose
`employees_master`	Employee data: role, branch, hire date, maturity status
`peer_baselines`	Expected mean/std per peer comparison metric, segmented by role × branch × window

Five Client Archetypes

Archetype	Deposit pattern	Dominant ops	Timing	Predictability
Salaried	Regular monthly deposits, stable amounts	versement + retrait	Morning hours	High
Small business	Irregular, varied amounts	All four ops	Random	Low
Student	Small irregular deposits	retrait-dominant	Evening / lunch	Medium
Retiree	Fixed monthly pension	retrait-dominant	Morning	Highest
Big business	Same as small biz, larger magnitude	All four ops	Random	Low

Cross-cutting structural flags (layered on archetypes, not additional archetypes):

- **PPE flag** (Personne Politiquement Expos'ee) — legal/political exposure status. Not a primary detector; acts as an alert-tier amplifier.
- **Multi-role flag** — client holds multiple economic identities (e.g., g'erant of multiple companies). Captures structural complexity, not demographic profiling.

Cold-start defense (three layers):

1. Regulatory rules and LAB/FT thresholds (10,000 DT reporting limit, near-threshold ratio in the 8,000–9,999 DT smurfing band)
2. Archetype baseline as proxy profile
3. Employee profile via dual-actor model

Model Design

LSTM — Next-step Prediction (multi-task head)

The LSTM takes the last 50 events of a client's history as input and predicts what the *next* event should look like. An event that deviates significantly from the prediction is flagged as anomalous.

Seven prediction targets:

- Operation type (softmax over 4 ops)
- Amount bucket (softmax, **including a dedicated 8–10k DT smurfing band**)
- Channel (softmax)
- Four binary *known-vs-new* flags: branch, employee, beneficiary, emitter

Strict architectural separation: the LSTM produces statistical anomaly scores only. LAB/FT regulatory thresholds live in the **Decision Service**, not in the model. This separation ensures that regulatory rules can be updated independently of the model.

Autoencoder — Reconstruction-based Detection

Trained on normal transactions only. An event with high reconstruction error is flagged as anomalous. Complements the LSTM: the autoencoder catches point anomalies (single unusual transactions) while the LSTM catches sequential anomalies (unusual patterns in transaction history).

Score Fusion and SHAP

Both model scores are fused in the Scoring Service. SHAP explanations are computed at scoring time and embedded in the payload, enabling the Compliance Dashboard to display human-readable explanations (e.g., "This versement is suspicious because the amount is 4x the client's average, it is the 3rd deposit in 48h, and the amount is a round multiple of 1,000 DT").

Synthetic Data Strategy

Generator Architecture — Option C (Calibrated Synthetic)

Three layers at the client level:

- **Layer 1 — Archetype prior:** aggregate distributions per archetype (e.g., "salaried clients average 1.2 versements/month of mean 850 DT")
- **Layer 2 — Per-client personality:** parameters drawn from the archetype prior, fixed at client creation (e.g., "client 4781 does 1.5 versements/month of mean 920 DT")
- **Layer 3 — Per-event noise:** each event is sampled from the client's personality with random variation

Architecture designed in-house; parameters calibrated from public sources (PaySim, Amen Bank annual report, CTAF).

Two distinct tools

- **Synthetic data pipeline** (Faker / SDV): produces the training dataset for the models
- **Live event simulator**: produces a real-time stream for demo and evaluation purposes

These serve different purposes and should not be conflated.

Source Extractions

v2 Extraction Template

Seven trust dimensions: recency, authority, peer review, **information volume** (renamed from “sample size”), geographic relevance, data type, reproducibility. Each scored Strong / Medium / Weak.

9-dimension coverage matrix (upgraded from v1’s 8 dimensions):

1. Amount distribution
2. Operation type mix
3. Transaction frequency
4. Inter-event time
5. Temporal patterns
6. Spatial / branch patterns
7. Channel mix
8. Beneficiary / emitter known-rate
9. **Operation sequence / transition patterns** (new — critical for the LSTM)

Each cell scored High / Medium / Low / Not covered.

Anomaly typology split into two sub-sections:

- **Behavioral Shapes** (the *form* of suspicious activity: smurfing, third-party deposits, etc.)
- **Institutional Framing** (how the institution categorizes it: LAB vs FT, predicate offense, etc.)

Operation mapping split into two modes:

- **Direct** (source records actual transactions matching our schema)
- **Indirect** (source describes patterns/typologies rather than raw transactions)

Source S1 — PaySim

Trust: 2 Strong / 4 Medium / 1 Weak
Primary role: beneficiary-graph-structure source
Secondary role: amount distribution shape + day-of-month seasonality
Operation mapping: retrait ↔ CASH-OUT, versement ↔ CASH-IN, virement ↔ TRANSFER; **ch‘eque not covered** (largest gap)
Anomaly base rate: $ $0.13% retained as initial calibration anchor
Key methodological insight: internal measurement quality and contextual transferability are scored independently (PaySim has good internal quality but limited transferability to the Tunisian banking context)

Source S2 — Amen Bank 2023 Annual Report

Trust: two-tier structure (Tier 1 = audited financials by BDO Tunisie + La G'en'erale d’Audit et Conseil; Tier 2 = unaudited management narrative)
Role: baseline / normal-population calibration (who are the clients, what is the bank’s infrastructure)

Key extractions:

- Client deposit composition: 6% institutional / 33% private businesses / 61% individuals
- Employee structural priors: 1,139 employees, 85.3% management ratio, 60.5% network-assigned, 61/39 gender split
- Branch network: 150 branches across 9 geographic zones, 5 Business Centers, 11 Self-Service Areas
- Multi-channel landscape: Amen Mobile, AmenPay, Amennet, teller, GAB, Western Union
- Confirmed existence of an already-deployed AI-AML system at the bank

Source S3 — CTAF 2021 Annual Report

Role: anomaly / flagged-population calibration (what do suspicious operations look like)
Template version: v2 (with all four patches applied)
Extraction strategy: two-pass (structural inventory of the full 1712-line document, then detailed extraction of the 4 typological case studies)

Calibration distinction: Amen Bank describes the *normal population* ($>99%*legitimateoperations*);*CTAFdesc* *flaggedpopulation* ∗ (<$1% that should fire the alarm). The synthetic generator needs both. Note: CTAF data reflects banks' notification behavior, not underlying criminal behavior.

Case 1 — Terrorism Financing (p. 15–17)

Actors: Subject X (Tunisian, imam, multi-role: g'erant of a jardin d'enfants + soci'et'e de commerce + association founder). Third-party depositors Y and Z (linked to extremist organizations). Foreign transferor from XLand.

Pattern: Slow drip over 10 years — repeated cash deposits by third parties (84,630 TND cumulative), sporadic foreign incoming wires with implausible declared motive ("aide familiale"), full cash conversion (retrait ≈ versement). Profile-flux inconsistency: imam profile vs. observed financial activity.

CTAF outcome: CNLCT listing (sanctions-equivalent), transmission to Parquet.

Key signals for our system: `is_third_party_deposit`, `lifetime_cash_conversion_ratio $\approx$ 1.0`, `is_foreign_inbound_implausible_motive`, archetype baseline z-score deviation.

Schema impact: `depositor_id` on versement payload confirmed as load-bearing; `lifetime_*` aggregates needed at client level (not per-account).

Case 2 — Money Mule Network (p. 18–19)

Actors: X (Franco-Tunisian-Swiss-British, non-resident) and Y (Anglo-Lebanese), married couple in L-Land. Multiple shell companies with 1 GBP capital.

Pattern: Three classic laundering stages — placement (foreign incoming wires, 3.5 MD enjeu), layering (assurance-vie 2.9 MD broken after 1 year, bons de caisse 2.4 MD cycled every 3 months), integration (clinic shares 1.5 MD, real estate 300 KD, Land Rover). Multi-account, multi-currency. Y → X inter-account consolidation.

CTAF outcome: Asset freeze + transmission to Procureur + judicial inquiry.

Scope note: the layering/integration phases (investment products) are **out of scope** for our 4-op system. Our system catches the placement-phase signals (multiple foreign wires, multi-account activity, Y → X consolidation). This is documented as an honest scope limitation; for the soutenance, the full pattern is presented with explicit marking of the covered subset.

Schema impact: `transit_velocity_*`, `multi_currency_activity_ratio_*`, `cross_jurisdiction_outflow_d` added to client profile. `sector_code` added to `accounts` table (not client profile — sector is an account-level attribute).

Case 3 — Corruption + Environmental Crime (p. 20–21)

Actors: X (PPE, g'erant of ALPHA — scrap metal recycling) and Y (PPE, administrative official dismissed 20 days before the trigger event). Z (straw man for fictitious real estate cover).

Pattern: PPE-to-PPE flow without economic background — 2 cheques of 50,000 DT each (round amounts, just under 100k threshold) from ALPHA's corporate account to Y's personal account. Y immediately disposes via cash withdrawals + cheques to unrelated parties. Reverse flow Y → ALPHA. Fictitious "promesse de vente" as cover. ALPHA obtained its battery-recycling licence from Y's administration in 2017 (the corruption link).

CTAF outcome: Both PPEs arrested + transmission + judicial inquiry.

PPE-flag validation: this case directly tests the design decision to treat PPE as a **layered flag** rather than a separate archetype. Result: the decision holds. Y's baseline is "salaried" — the anomaly is caught by archetype-baseline z-score deviation; the PPE flag amplifies the alert tier. The flag is a severity multiplier, not a feature primary.

Schema impact: `controlled_accounts_list` (minimal BO graph), `cash_disposal_velocity_*`, `reverse_flow_ratio_90d`. Enriched event gains `is_ppe_to_ppe_flow`, `is_corporate_to_personal_same_contr` `is_round_amount_under_threshold`.

Case 4 — Trade-Based Money Laundering (p. 22–23)

Actors: X (Franco-Tunisian, declared profession: engineer, undisclosed: g'erant-associ'e of BETA, a medical-dental import company). Multiple physical-person depositors/transferors. Foreign medical-dental supplier companies.

Pattern: KYC profile vs. observed activity mismatch — personal EUR account used as commercial conduit. 342k EUR total inflow (138k cash + 204k transfers from multiple physical persons), 282k EUR outflow to foreign medical-dental companies. CTAF cross-checked via TTN (Tunisia Trade Net): the foreign companies receiving X's personal transfers are the same companies BETA imports from. Sub-invoicing of imported goods.

CTAF outcome: Transmission to Procureur + judicial inquiry.

Schema impact: `declared_vs_observed_consistency_score` (new KYC batch job, weekly cadence), `inflow_emitter_type_distribution`, `multi_physical_person_inflow_count_90d`. TTN cross-system integration documented as out-of-scope / production roadmap.

Architectural Decisions

#	Decision	Rationale
D1	Tagged-union raw event envelope	Preserves cross-operation temporal ordering for the LSTM
D2	Hot/cold profile + dual-actor model	Separates real-time (Redis) and history (PostgreSQL); models client and employee independently
D3	SHAP co-located in Scoring Service	Simplifies Dashboard (single topic in)
D4	RedPanda over Kafka	Compatible API, lighter runtime
D5	Calibrated synthetic data (Option C)	In-house architecture, parameters from public sources
D6	LSTM sequence length = 50	Matches Redis inference buffer; prevents train/inference drift
D7	v2 template: 3 patches + 9th coverage dim	Patches identified in S2; sequence cell revealed by CTAF
D8	7-field per-case extraction template	Standardizes case extraction for v2 feeding
D9	3-layer amendment principle	Raw → Profile → Enriched: changes propagate and must be checked
D10	Structural flags (PPE, multi-role)	Cross-cutting risk attributes layered on archetypes; validated by Case 3
D11	`depositor_id` enforcement on versement	Third-party deposit pattern (Case 1) requires it
D12	`lifetime_*` aggregates at client level	Per-holder, not per-account, to catch multi-account patterns
D13	Scope kept at 4 caisse operations	Investment products out of scope; documented as roadmap
D14	`sector_code` on `accounts` table	Sector is an account-level attribute, not person-level
D15	Minimal viable controller graph	`controlled_accounts_list` on PPE clients; extensible later
D16	KYC consistency batch job	Declared vs. observed profile mismatch is a slow-tempo signal
D17	Counterparty features in enriched event	Section 12: counterparty archetype/sector/flags when internal

#	Decision	Rationale
D18	Threshold bands deferred to S4	Await BCT circulars for exhaustive threshold enumeration

Schema Amendments Pending Consolidation

Identified during CTAF extraction. To be formally applied to the checkpoint schemas next session.

Layer 1 — Raw Event

- `versement.payload.depositor_id`: enforcement strengthened (required when depositor ≠ holder)
- Verify: `remise_cheques.payload` should expose `drawer_account_type` and `drawer_account_id`
- Verify: `virement.payload` should expose `counterparty_country`

Layer 2 — Client Profile (new fields by source case)

Case 1: `lifetime_versement_sum`, `lifetime_versement_count`, `lifetime_retrait_sum`, `lifetime_retrait_count`, `lifetime_virement_in_sum` (domestic/international split), `lifetime_third_party_d` `lifetime_cash_conversion_ratio`

Case 2: `account_count_active_90d`, `account_count_active_lifetime`, `multi_currency_activity_ratio_30` `cross_jurisdiction_outflow_distribution`, `transit_velocity_p50_30d/_p95_30d` (and 180d)

Case 3: `controlled_accounts_list` (minimal BO graph), `cheque_drawer_distribution`, `cash_disposal_velocity_p50/_p95`, `reverse_flow_ratio_90d`

Case 4: `declared_vs_observed_consistency_score` (batch job), `inflow_emitter_type_distribution`, `outflow_destination_sector_distribution`, `multi_physical_person_inflow_count_90d`

Layer 3 — Enriched Event

New boolean rule signals: `is_third_party_deposit`, `is_foreign_inbound_implausible_motive`, `is_high_velocity_transit`, `is_multi_currency_client`, `is_cross_jurisdiction_concentrated_outflow`, `is_ppe_to_ppe_flow`, `is_ppe_to_ppe_via_controlled_account`, `is_corporate_to_personal_same_controlle` `is_round_amount_under_threshold`, `is_immediate_cash_disposal`, `is_reverse_flow_to_recent_source`, `is_kyc_inconsistency`, `is_personal_account_commercial_outflow`, `is_multi_physical_person_fan_in`, `is_personal_foreign_currency_sector_concentration`

New section 12: counterparty-side derived features

Batch Layer

- `archetype_baselines` segmented by sector × archetype × flag-set
- Coarse sector taxonomy ($ $20 buckets) to be enumerated next session
- KYC consistency batch job (weekly cadence)

Cold Store

- `accounts` table: add `sector_code` (coarse categorical)
- Verify `currency` and `account_type` columns

Next Milestones

Immediate:

1. Complete CTAF structural extractions (instruments %, financial tier distribution, BBE concentration, predicate taxonomy, geographic cartography, encodable rules)
2. Consolidated S3 LaTeX checkpoint
3. Enumerate the coarse sector taxonomy ($ $20 buckets) for `accounts.sector_code`
4. Formally apply schema amendments to Checkpoints 1, 3.2, 3.3, 3.4

Next source:

5. S4 — BCT circulars: `2018-09` (internal control LAB/FT), `2018-07` (manual exchange), `2021-05` (governance). Exhaustive regulatory threshold enumeration.

Deferred / production roadmap:

6. TTN integration (cross-system TBML detection — Case 4)
7. Investment product channel extension (Case 2)